TUTORIAL 9

Linear Maps and Matrix Representations

MOTIVATION AND GUIDING IDEA

Motivation. A representing matrix is a computational tool that depends on the ordered bases of the domain and codomain. Changing coordinates leaves the map unchanged but can greatly simplify repeated calculations. Always identify the input and output coordinate systems to avoid reversing a change of basis.

NOTATION AND CONVENTIONS

- For ordered bases $B = (u_1, \dots, u_n)$ of U and $C = (v_1, \dots, v_m)$ of V, the notation $[T]_C^B$ denotes an $m \times n$ matrix taking B-coordinates as input and returning C-coordinates.
- $S \circ T$ applies T first and S second; I_V is the identity map on V.
- $P_{B \leftarrow B'} = [I_V]_B^{B'}$ takes $[v]_{B'}$ to $[v]_B$. The arrow indicates the direction of coordinate conversion.
- If B is an ordered basis of $\mathbb{F}^n$ and E is the standard basis, $S_B = [\, b_1 \ \cdots \ b_n \,] = [I]_E^B$ is the basis-column matrix. Hence $[v]_E = S_B [v]_B$.
- $[T]_B$ abbreviates $[T]_B^B$ for an operator $T : V \to V$. Square matrices A, D are similar if $D = P^{-1}AP$ for an invertible P.

DEFINITION 9.1

Let U, V be finite-dimensional vector spaces over the same field, with ordered bases B, C, and let $T : U \to V$ be linear. Its representing matrix is defined by its columns:

$$[T]_C^B = \big[\, [T(u_1)]_C \ \cdots \ [T(u_n)]_C \,\big], \qquad [T(u)]_C = [T]_C^B [u]_B.$$

For another ordered basis B' of the domain, the columns of $P_{B \leftarrow B'}$ are the B-coordinate vectors of the vectors of B'. Its inverse is $P_{B' \leftarrow B}$. If $R : V \to W$ is linear and D is an ordered basis of W, then

$$[R \circ T]_D^B = [R]_D^C [T]_C^B.$$

If T is an isomorphism, $[T^{-1}]_B^C = ([T]_C^B)^{-1}$. For an operator with standard matrix A, the matrix in the ordered basis B is $S_B^{-1} A S_B$. Domain and codomain bases must match at every matrix product.

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MOTIVATION AND GUIDING IDEA

Construct a map from prescribed vector images, check compatibility with linear dependence relations, compute a representing matrix and change bases. The emphasis is on the connection between an abstract map, coordinates and matrix calculations.

9.1 Determining a map from vector images

PROPOSITION 9.1

Let V, W be vector spaces over the same field $\mathbb{F}$, let $B = (v_1, \dots, v_n)$ be an ordered basis of V, and let $w_1, \dots, w_n \in W$ be arbitrary vectors, not necessarily independent. There is a unique linear map $T : V \to W$ with $T(v_i) = w_i$ for every i. Explicitly, if $v = \sum_{i=1}^{n} a_i v_i$, then

$$T(v) = \sum_{i=1}^{n} a_i w_i.$$

The formula is well defined because basis expansions are unique. It is linear because coordinates of sums and scalar multiples are computed entry by entry. Any linear map with the prescribed images must obey this formula, proving uniqueness. In particular, between any two vector spaces over the same field there is at least one linear map: the zero map.

EXERCISE 9.1

(a) $T(1,0,0)^T = (2,0,1)^T$, $T(1,1,0)^T = (1,1,0)^T$, $T(1,0,1)^T = (0,2,1)^T$;

(b) $T(2,1,4)^T = (1,2,0)^T$, $T(1,0,2)^T = (1,-1,1)^T$, $T(0,1,0)^T = (-1,4,0)^T$.

Please decide in each case whether a linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$ satisfies the requirements. If it exists, please find an explicit formula.

WORKED SOLUTION

In part (a), the three inputs form a basis, so the map exists uniquely. Computing the images of the standard basis gives the formula. In part (b), the proposed outputs do not preserve the displayed input relation, so no such map exists.

(a) $T(e_1) = (2,0,1)^T$, $T(e_2) = (1,1,0)^T - T(e_1) = (-1,1,-1)^T$,

$T(e_3) = (0,2,1)^T - T(e_1) = (-2,2,0)^T$, $T(x,y,z)^T = (2x - y - 2z, y + 2z, x - y)^T$.

(b) $(2,1,4)^T = 2(1,0,2)^T + (0,1,0)^T$,

$2(1,-1,1)^T + (-1,4,0)^T = (1,2,2)^T \neq (1,2,0)^T$.

MOTIVATION AND GUIDING IDEA

Existence criterion for prescribed images. The requirements $T(u_i) = w_i$ can be satisfied by a linear map if and only if, for every choice of coefficients a_i,

$$\sum_i a_i u_i = 0 \quad \Longrightarrow \quad \sum_i a_i w_i = 0.$$

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Under this condition, define a map on $U = \operatorname{span}\{u_i\}$ by $T(\sum_i a_i u_i) = \sum_i a_i w_i$. The difference of two representations is a dependence relation, so both yield the same image. Linearity follows from the definition. If the domain is finite-dimensional, extend a basis of U to a basis of V, and freely prescribe images of the extra basis vectors. The map is unique if $U = V$. If $U \neq V$ and $W \neq \{0\}$, two different images may be chosen for an extra basis vector, so the extension is not unique. If the u_i already form a basis, there are no nontrivial dependence relations, and every choice of images defines a unique linear map.

EXERCISE 9.2

$$T : \mathbb{R}^2 \to \mathbb{R}_3[X], \qquad T((1,0)^T) = 1 + 2X^2, \quad T((1,1)^T) = 4 + X + 2X^2.$$

Please find the unique linear map satisfying these requirements.

WORKED SOLUTION

The two given inputs form a basis. Linearity forces the formula, and substitution reproduces both prescribed outputs.

$$T((0,1)^T) = T((1,1)^T) - T((1,0)^T) = 3 + X,$$

$$T((x,y)^T) = x(1 + 2X^2) + y(3 + X) = (x + 3y) + yX + 2xX^2.$$

9.2 Composition of maps

RECALL AND TOOLS

- Let U, V, W be vector spaces over a field $\mathbb{F}$.
- Let $T : U \to V$ and $S : V \to W$ be linear maps.
- The map $S \circ T : U \to W$, defined for $u \in U$ by $(S \circ T)(u) = S(T(u))$, is their **composition**.
- $S \circ T$ is linear over $\mathbb{F}$.
- If both maps are surjective, their composition is surjective. If both are injective, their composition is injective. In particular, a composition of isomorphisms is an isomorphism.

EXERCISE 9.3

$$T : \mathbb{R}^2 \to M_2(\mathbb{R}), \qquad T(a,b)^T = \begin{pmatrix} 2a & a-b \\ a+b & 3b \end{pmatrix},$$

$$S : M_2(\mathbb{R}) \to \mathbb{R}_3[X], \qquad S\begin{pmatrix} q & r \\ s & t \end{pmatrix} = qX^2 + (r - 2s)X + t.$$

Please compute $S \circ T$.

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EXERCISE 9.3 CONTINUED

WORKED SOLUTION

Apply T first and substitute its four matrix entries into the formula for S.

$$(S \circ T)(a,b)^T = (2a)X^2 + \big((a-b) - 2(a+b)\big)X + 3b = 2aX^2 - (a+3b)X + 3b.$$

9.3 Representing a linear map by a matrix

DEFINITION 9.2

Let U be an n-dimensional vector space over $\mathbb{F}$ with ordered basis $B = (u_1, \ldots, u_n)$.
Let V be an m-dimensional vector space over the same field with ordered basis $C = (v_1, \ldots, v_m)$.
For a linear map $T : U \to V$, define the matrix

$$[T]_C^B = \begin{pmatrix} | & | & | & | \\ [T(u_1)]_C & [T(u_2)]_C & \cdots & [T(u_n)]_C \\ | & | & | & | \end{pmatrix} \in \mathbb{F}^{m \times n}$$

$[T]_C^B$ is the **matrix of T relative to the ordered bases B and C.**

Construction

First compute $T(u_i)$, the images of the basis vectors of the domain.

Then express each image in coordinates relative to C, and use these coordinate columns in the given order.

PROPOSITION 9.2

Let U, V be vector spaces over $\mathbb{F}$, with ordered bases B of U and C of V. For a linear map $T : U \to V$ and every $u \in U$,

$$[T(u)]_C = [T]_C^B \cdot [u]_B$$

Compute this matrix once for the chosen ordered bases. It then converts the coordinate vector of any new input into the coordinate vector of its image relative to C.

EXERCISE 9.4

$$V = M_3(\mathbb{R}), \qquad T(A) = 2(A - A^T),$$
$$U = \{A : A = A^T\}, \qquad W = \{A : A = -A^T\}.$$

Please prove linearity, please find the kernel and image and their dimensions, and please give a representing matrix in a basis formed by a basis of U followed by a basis of W.

WORKED SOLUTION

Here E_{ij} has a single 1 in position (i,j) and zeros elsewhere. The free matrix entries give the displayed bases. The map vanishes on symmetric matrices and multiplies skew-symmetric

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EXERCISE 9.4 CONTINUED

matrices by 4, giving the diagonal representation.

$$T(\alpha A+\beta B)=2\big(\alpha A+\beta B-\alpha A^T-\beta B^T\big)=\alpha T(A)+\beta T(B).$$

$$T(A)=0 \iff A=A^T, \qquad \ker T=U.$$

$$T(A)^T=-T(A), \qquad K\in W \implies T(K/4)=K.$$

$$\operatorname{Im} T=W, \qquad B_U=(E_{11},E_{22},E_{33},E_{12}+E_{21},E_{13}+E_{31},E_{23}+E_{32}),$$

$$B_W=(E_{12}-E_{21},E_{13}-E_{31},E_{23}-E_{32}), \qquad \dim U=6, \quad \dim W=3, \quad 6+3=9.$$

$$A=\frac{A+A^T}{2}+\frac{A-A^T}{2}\in U+W, \quad U\cap W=\{0\}, \qquad [T]_{(B_U,B_W)}=\operatorname{diag}(0,0,0,0,0,0,4,4,4).$$

EXERCISE 9.5

$$[T]_E=A=\begin{pmatrix}0&2&-1\\0&3&1\\0&0&4\end{pmatrix}, \qquad V=\mathbb{R}^3.$$

Please find nonzero vectors v,u satisfying $T(v)=0$ and $T(u)=4u$. Then please find w satisfying $T(w)=w+u$.

WORKED SOLUTION

The first column is zero, so e_1 lies in the kernel. Solving the second system gives an eigenvector for 4. Scaling that eigenvector by $\frac{1}{3}$ yields the last equation.

$$v=(1,0,0)^T, \qquad Av=0.$$

$$(A-4I)u=0 \iff -4u_1+2u_2-u_3=0, \quad -u_2+u_3=0.$$

$$u=(1,4,4)^T, \qquad Au=(4,16,16)^T=4u.$$

$$w=\tfrac{1}{3}u=(\tfrac{1}{3},\tfrac{4}{3},\tfrac{4}{3})^T, \qquad T(w)=\tfrac{4}{3}u=w+u.$$

EXERCISE 9.6

Let $V=\mathbb{C}[x]$, the vector space of all complex polynomials.
Define the linear operator $N:V\to V$ by

$$(Np)(x):=-(7-x)\,p'(x).$$

For each $\theta\in\mathbb{R}$, define $U_\theta:V\to V$ by

$$U_\theta:=\exp(-i\theta N^2)=\sum_{k=0}^{\infty}\frac{(-i\theta)^k}{k!}N^{2k}.$$

Please use the following two ordered bases:

$$E=(1,x,x^2,x^3,\dots), \qquad B=(7,\ (7-x),\ (7-x)^2,\ (7-x)^3,\dots).$$

1. Please prove that N and U_θ are linear.

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EXERCISE 9.6 CONTINUED

2. For $n \geq 1$, set $b_n(x) = (7-x)^n$, and set $b_0(x) = 7$. Please compute $N(b_n)$ and $N^2(b_n)$.
3. Please compute $U_\theta(b_n)$ for all $n \geq 0$ and please deduce the form of $[U_\theta]_B$.
4. Please find the matrix of U_θ restricted to $P_3 = \text{span}\{1, x, x^2, x^3\}$ in the ordered basis $E_3 = (1, x, x^2, x^3)$.

WORKED SOLUTION

1. Differentiation is linear, so

$$N(\alpha p + \beta r) = -(7-x)(\alpha p' + \beta r') = \alpha N(p) + \beta N(r).$$

To justify the series, restrict to P_d, the polynomials of degree at most d. This space is invariant under N, so on P_d the series is the exponential series of a finite matrix, converging for every θ. These definitions agree as d increases and therefore define a map on $\mathbb{C}[x]$. Linearity follows by taking limits of the partial sums in each coordinate.

2. For $n \geq 1$,

$$b_n'(x) = \frac{d}{dx}(7-x)^n = n(7-x)^{n-1} \cdot (-1) = -n(7-x)^{n-1}.$$

so

$$N(b_n) = -(7-x)\, b_n'(x) = -(7-x)\big(-n(7-x)^{n-1}\big) = n(7-x)^n = nb_n.$$

and hence

$$N^2(b_n) = N(nb_n) = nN(b_n) = n \cdot nb_n = n^2 b_n.$$

Also $N(b_0) = N(7) = 0$, so $N^2(b_0) = 0$.

3. Since $N^{2k}(b_n) = n^{2k} b_n$ for $n \geq 1$, the series gives

$$U_\theta(b_n) = \sum_{k=0}^{\infty} \frac{(-i\theta)^k}{k!}\, n^{2k}\, b_n = \exp(-i\theta n^2)\, b_n.$$

For $n = 0$, $U_\theta(b_0) = b_0$.

Thus the matrix in B is diagonal:

$$[U_\theta]_B = \text{diag}\big(1,\ e^{-i\theta},\ e^{-i4\theta},\ e^{-i9\theta}, \dots\big).$$

4. Write $q_1 = e^{-i\theta}$, $q_4 = e^{-i4\theta}$ and $q_9 = e^{-i9\theta}$. On P_3, use b_0, b_1, b_2, b_3. Their images are $b_0, q_1 b_1, q_4 b_2, q_9 b_3$, respectively.

Expand these four polynomials in E_3:

$$b_0 = 7, \qquad b_1 = 7 - x, \qquad b_2 = (7-x)^2 = x^2 - 14x + 49,$$

$$b_3 = (7-x)^3 = -x^3 + 21x^2 - 147x + 343.$$

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EXERCISE 9.6 CONTINUED

Let C be the matrix with coordinate columns $[b_0]_{E_3}, [b_1]_{E_3}, [b_2]_{E_3}, [b_3]_{E_3}$:

$$C = \begin{pmatrix} 7 & 7 & 49 & 343 \\ 0 & -1 & -14 & -147 \\ 0 & 0 & 1 & 21 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

This matrix is invertible. The formula $[U_\theta]_{E_3} = C\,\mathrm{diag}(1, q_1, q_4, q_9)C^{-1}$ gives the following matrix:

$$[U_\theta]_{E_3} = \begin{pmatrix} 1 & 7(1-q_1) & 49(1-2q_1+q_4) & 343(1-3q_1+3q_4-q_9) \\ 0 & q_1 & 14(q_1-q_4) & 147(q_1-2q_4+q_9) \\ 0 & 0 & q_4 & 21(q_4-q_9) \\ 0 & 0 & 0 & q_9 \end{pmatrix}.$$

EXERCISE 9.7

$$T : \mathbb{R}^3 \to \mathbb{R}^2, \qquad T(x,y,z)^T = (3x+2y-z, 2x-y+4z)^T.$$

Please compute the standard-basis matrix using the images of the basis vectors.

WORKED SOLUTION

The basis-vector images are the matrix columns. The final multiplication verifies the formula for every input.

$$T(e_1) = (3,2)^T, \quad T(e_2) = (2,-1)^T, \quad T(e_3) = (-1,4)^T,$$

$$[T] = \begin{pmatrix} 3 & 2 & -1 \\ 2 & -1 & 4 \end{pmatrix}, \qquad [T]\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3x+2y-z \\ 2x-y+4z \end{pmatrix}.$$

EXERCISE 9.8

$$T : \mathbb{R}_2[X] \to \mathbb{R}^2, \qquad T(p) = (p(0), p(-1))^T,$$
$$B = (1, 2+X, 1+X+2X^2), \qquad C = ((1,-1)^T, (0,1)^T).$$

Please compute $[T]_C^B$ and please verify $[T(p)]_C = [T]_C^B[p]_B$ for $p = a + bX + cX^2$.

WORKED SOLUTION

Compute each image and convert it to C-coordinates. To verify the identity, first equate coefficients in the B-expansion and then multiply by the matrix.

$$[(r,s)^T]_C = (r, r+s)^T,$$

$$T(1) = (1,1)^T, \quad T(2+X) = (2,1)^T, \quad T(1+X+2X^2) = (1,2)^T,$$

$$M = [T]_C^B = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \end{pmatrix}.$$

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EXERCISE 9.8 CONTINUED

$$p = \alpha + \beta(2+X) + \gamma(1+X+2X^2) \implies \gamma = c/2, \quad \beta = b - c/2, \quad \alpha = a - 2b + c/2.$$

$$[p]_B = \begin{pmatrix} a - 2b + c/2 \\ b - c/2 \\ c/2 \end{pmatrix}, \qquad M[p]_B = \begin{pmatrix} a \\ 2a - b + c \end{pmatrix}.$$

$$T(p) = (a, a - b + c)^T, \qquad [T(p)]_C = (a, 2a - b + c)^T.$$

9.3.1 Change-of-coordinate matrices

RECALL AND TOOLS

Definition: Let V be a finite-dimensional vector space over $\mathbb{F}$, with ordered bases B, B'. The **change-of-coordinate matrix** $[I_V]_B^{B'}$ from B' to B is the representing matrix $[T]_B^{B'}$ for the **identity map** $T : V \to V$, $T(v) = v$ for every $v \in V$.

- The fundamental identity is $[v]_B = [I_V]_B^{B'}[v]_{B'}$.
- The identity map is denoted I_V, with $I_V(v) = v$.
- Its columns are the coordinate vectors of the vectors of B' relative to the ordered basis B.

EXERCISE 9.9

$$B = (1+X, 2+3X), \qquad B' = (3+4X, 1+2X), \qquad p = a + bX.$$

Please compute the transition matrix $[I]_B^{B'}$ and please verify $[p]_B = [I]_B^{B'}[p]_{B'}$.

WORKED SOLUTION

The basis matrices contain polynomial coefficients in $(1, X)$. The transition first converts to standard coordinates and then to B. The coordinate formulas and final product explicitly verify the identity.

$$S_B = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad S_{B'} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}, \quad S_B^{-1} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix},$$

$$[I]_B^{B'} = S_B^{-1} S_{B'} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

$$[p]_B = \begin{pmatrix} 3a - 2b \\ -a + b \end{pmatrix}, \qquad [p]_{B'} = \begin{pmatrix} a - b/2 \\ -2a + 3b/2 \end{pmatrix},$$

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a - b/2 \\ -2a + 3b/2 \end{pmatrix} = \begin{pmatrix} 3a - 2b \\ -a + b \end{pmatrix}.$$

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9.3.2 Matrices of compositions and inverse maps

PROPOSITION 9.3

Let U, V, W be finite-dimensional vector spaces over $\mathbb{F}$, with ordered bases A, B, C, respectively.
Let $S : U \to V$ and $T : V \to W$ be linear. Then

$$[T \circ S]_C^A = [T]_C^B [S]_B^A$$

Corollary

If $T : V \to W$ is an **isomorphism** and B, C are ordered bases of V, W, respectively, then

$$[T^{-1}]_B^C = \left([T]_C^B\right)^{-1}$$

9.4 Similar matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are similar if there is an invertible $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

1. $$\det(A) = \det(B)$$
2. $$\operatorname{trace}(A) = \operatorname{trace}(B)$$
3. $$\operatorname{rank}(A) = \operatorname{rank}(B)$$

EXERCISE 9.10

$$A, B \in \mathbb{F}^{n\times n}.$$

Please prove or disprove: if A, B are similar, then A^k, B^k are similar for every $k \geq 1$; if A^k, B^k are similar for some $k \geq 1$, then A, B are similar.

WORKED SOLUTION

The first claim is true: cancelling adjacent inverse factors gives similarity through the same P. The second claim is false over every field. The displayed cubes agree, but a matrix similar to zero must be zero, whereas B is nonzero.

$$B = P^{-1}AP \quad \Longrightarrow \quad B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

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EXERCISE 9.10 CONTINUED

$$A = 0_{3\times 3}, \qquad B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

$$A^3 = B^3 = 0, \qquad B^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad B \neq 0.$$

EXERCISE 9.11

$$T : \mathbb{R}^2 \to \mathbb{R}^2, \quad T(x,y)^T = (2x+y, x-y)^T, \qquad B = ((1,1)^T, (1,-2)^T).$$

Please compute $[T]_E^B$, please find v with $T(v) = (9,-6)^T$, and please check the result in both coordinate systems.

WORKED SOLUTION

The columns are the images in standard output coordinates. Dividing by 3 gives the input B-coordinates, and rebuilding in B gives the vector itself.

$$T(1,1)^T = (3,0)^T, \qquad T(1,-2)^T = (0,3)^T, \qquad [T]_E^B = 3I_2.$$

$$[v]_B = (3,-2)^T, \qquad v = 3(1,1)^T - 2(1,-2)^T = (1,7)^T.$$

$$T(1,7)^T = (2+7, 1-7)^T = (9,-6)^T, \qquad 3I_2(3,-2)^T = (9,-6)^T.$$

EXERCISE 9.12 EXAM QUESTION

Please prove or disprove the following statements.

1. There do not exist fields $\mathbb{F}_1 \subseteq \mathbb{F}_2$ and a vector space V over $\mathbb{F}_2$, also regarded as an $\mathbb{F}_1$-vector space by restricting scalars, such that
$$\dim_{\mathbb{F}_1} V = 2 \dim_{\mathbb{F}_2} V.$$
2. If $A \in M_n(\mathbb{F})$ is invertible, then every $B \in M_{n\times m}(\mathbb{F})$ satisfies
$$N(AB) = N(B).$$
3. Let $n \geq 1$. For every integer k with $0 \leq k \leq n^2$, there exists $A \in M_n(\mathbb{R})$ with $\mathrm{tr}(A) = k$ that is similar only to itself.

WORKED SOLUTION

1. The claim is false. Take $\mathbb{F}_1 = \mathbb{R}$, $\mathbb{F}_2 = \mathbb{C}$ and $V = \mathbb{C}^2$. Appropriate bases are
$$B_{\mathbb{C}} = ((1,0),(0,1)), \qquad B_{\mathbb{R}} = ((1,0),(i,0),(0,1),(0,i)).$$
$$\dim_{\mathbb{C}} V = 2, \qquad \dim_{\mathbb{R}} V = 4 = 2 \dim_{\mathbb{C}} V.$$

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EXERCISE 9.12 EXAM QUESTION CONTINUED

2. True. Prove both inclusions. If $x \in N(B)$, then $Bx = 0$, and

$$ABx = A(Bx) = A0 = 0.$$

so $x \in N(AB)$. Hence $N(B) \subseteq N(AB)$. Conversely, if $x \in N(AB)$, then $ABx = 0$. Multiplication on the left by A^{-1} gives

$$Bx = (A^{-1}A)Bx = A^{-1}0 = 0.$$

so $x \in N(B)$, proving $N(AB) \subseteq N(B)$. Together the inclusions give the required equality.

3. True. Define

$$A = \frac{k}{n} I_n.$$

Since $n \geq 1$ and the field is $\mathbb{R}$, the quotient is defined, and

$$\operatorname{tr}(A) = \sum_{j=1}^{n} \frac{k}{n} = n\frac{k}{n} = k.$$

For every invertible $P \in M_n(\mathbb{R})$,

$$P^{-1}AP = P^{-1}\left(\frac{k}{n} I_n\right) P = \frac{k}{n} P^{-1}P = \frac{k}{n} I_n = A.$$

Thus every matrix similar to A equals A. In fact this construction works for every real k, including all the values in the statement.

Additional practice

EXERCISE 9.13 FOUNDATION

$$T(x, y) = (2x - y, x + y), \quad B = ((1, 0)^T, (1, 1)^T), \quad C = ((1, 1)^T, (0, 1)^T), \quad v = (3, 2)^T.$$

Please compute $[T]_C^B$ and $[T^{-1}]_B^C$. Please verify the coordinate identity and please recover v using the inverse.

WORKED SOLUTION

Convert the images of the B-vectors to C-coordinates and place them in the columns. The inverse takes C-coordinates and returns B-coordinates. Reconstruction in B recovers $(3, 2)^T$.

$$[(r, s)^T]_C = (r, s - r)^T, \quad T(1, 0)^T = (2, 1)^T, \quad T(1, 1)^T = (1, 2)^T.$$

$$M = [T]_C^B = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}, \quad \det M = 3, \quad [T^{-1}]_B^C = M^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}.$$

$$v = (1, 0)^T + 2(1, 1)^T, \quad [v]_B = (1, 2)^T, \quad T(v) = (4, 5)^T,$$

$$[T(v)]_C = (4, 1)^T = M(1, 2)^T, \qquad M^{-1}(4, 1)^T = \frac{1}{3}(3, 6)^T = (1, 2)^T.$$

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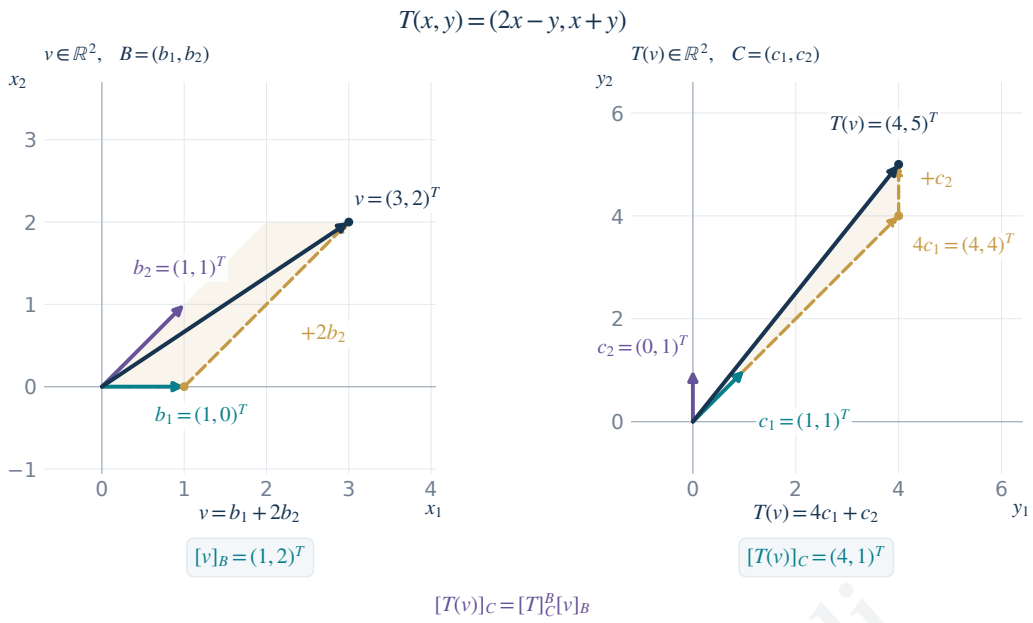


Figure 9.1. The arrows show vectors in standard coordinates; the gold paths show the decompositions $v = b_1 + 2b_2$ and $T(v) = 4c_1 + c_2$. Coordinate columns appear separately: $[v]_B = (1,2)^T$ and $[T(v)]_C = (4,1)^T$. The vector $T(v)$ itself is $(4,5)^T$.

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EXERCISE 9.14 CHALLENGE

$$A=\begin{pmatrix}4&2\\2&4\end{pmatrix},\quad B=((1,1)^T,(1,-1)^T),\quad x_{k+1}=Ax_k,\quad x_0=(2,1)^T.$$

Please compute $D=[T]_B$ for $T(x)=Ax$ and $P=[I]_E^B$. Please prove $A^k=PD^kP^{-1}$ for every $k\geq 0$ and please solve the recurrence.

WORKED SOLUTION

The columns of P are the new basis vectors in standard coordinates. Cancelling adjacent $P^{-1}P$ factors proves the power identity; for $k=0$ both sides are the identity. Induction on the recurrence gives $x_k=A^kx_0$. In the basis B, the two components are multiplied separately by 6 and 2.

$$A(1,1)^T=6(1,1)^T,\quad A(1,-1)^T=2(1,-1)^T,$$

$$D=\begin{pmatrix}6&0\\0&2\end{pmatrix},\quad P=\begin{pmatrix}1&1\\1&-1\end{pmatrix},\quad P^{-1}=\frac{1}{2}\begin{pmatrix}1&1\\1&-1\end{pmatrix},\quad A=PDP^{-1},$$

$$A^k=PD^kP^{-1}=\frac{1}{2}\begin{pmatrix}6^k+2^k&6^k-2^k\\6^k-2^k&6^k+2^k\end{pmatrix},$$

$$x_k=A^kx_0=\frac{1}{2}\begin{pmatrix}3\cdot 6^k+2^k\\3\cdot 6^k-2^k\end{pmatrix},\quad x_0=\frac{1}{2}(4,2)^T=(2,1)^T.$$

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EXERCISE 9.15 CHALLENGE

Let $\dim V = n$ over a field F, and let $T : V \to V$ be linear with $T^2 = 0$. Put $r = \operatorname{rank} T$. Please prove that $2r \le n$ and that some basis gives the matrix

$$\begin{pmatrix} 0 & I_r & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The three row and column block sizes are $r, r, n - 2r$. Blocks of size zero are omitted.

WORKED SOLUTION

The identity $T^2 = 0$ gives $\operatorname{Im} T \subseteq \ker T$. Rank-nullity yields $r \le n - r$, so $2r \le n$. Choose a basis $u_1, \dots, u_r$ of the image and extend it to a basis $u_1, \dots, u_r, z_1, \dots, z_{n-2r}$ of the kernel. For each i, choose w_i with $T(w_i) = u_i$. The list

$$\mathcal{B} = (u_1, \dots, u_r, w_1, \dots, w_r, z_1, \dots, z_{n-2r})$$

is independent. Applying T to a zero linear combination shows that all coefficients of the w_i vanish, because the u_i are independent. The remaining combination is in a basis of the kernel, so all its coefficients vanish too. The list has n vectors and is therefore a basis. In this basis $T(u_i) = T(z_j) = 0$ and $T(w_i) = u_i$. Writing these images as columns gives the displayed matrix. When $r = 0$, $T = 0$ and every basis works.

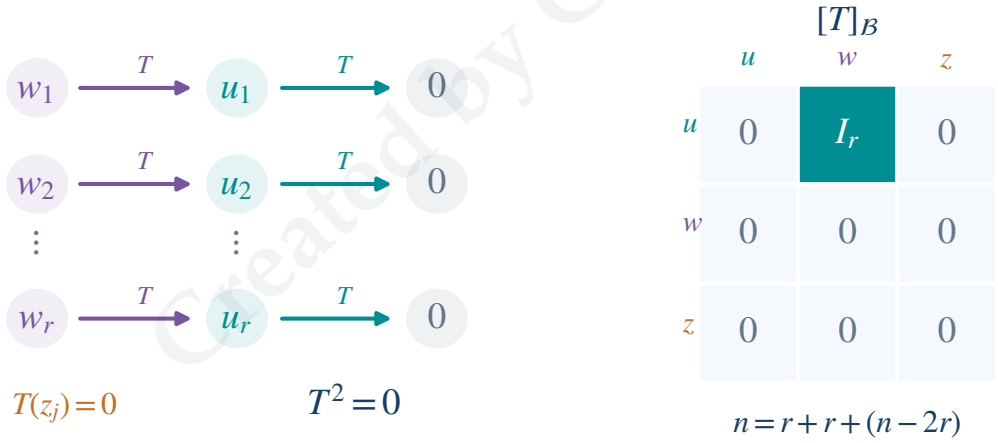


Figure 9.2. The schematic shows the case $r \ge 3$. Each w_i maps to u_i, and each u_i maps to zero. The vectors z_j complete $u_1, \dots, u_r$ to a basis of the kernel. The basis groups are ordered u, w, z.

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EXERCISE 9.16 CHALLENGE

Let $A \in M_n(F)$ have rank 1, where $n \geq 2$, and put $\alpha = \operatorname{tr} A$, the sum of the diagonal entries. Please prove that A is similar to

$$\begin{cases} \operatorname{diag}(\alpha, 0, \ldots, 0), & \alpha \neq 0, \\ E_{12}, & \alpha = 0, \end{cases}$$

where E_{12} has a single 1 in position $(1, 2)$. Please deduce that two rank-one matrices of the same size and trace are similar.

WORKED SOLUTION

Write $A = uv^T$ with $u, v \neq 0$ and define $\varphi(x) = v^T x$. Then

$$Ax = \varphi(x)u, \qquad \operatorname{tr} A = \sum_i u_i v_i = \varphi(u) = \alpha.$$

Since $\varphi \neq 0$, its kernel has dimension $n-1$. If $\alpha \neq 0$, then $u \notin \ker \varphi$. A basis $\mathcal{B} = (u, z_2, \ldots, z_n)$ obtained by adjoining u to a basis of the kernel satisfies $Au = \alpha u$ and $Az_j = 0$. Its representing matrix is $\operatorname{diag}(\alpha, 0, \ldots, 0)$. If $\alpha = 0$, then $u \in \ker \varphi$. Choose w with $\varphi(w) = 1$ and extend u to a basis $(u, z_3, \ldots, z_n)$ of the kernel. The list $\mathcal{B} = (u, w, z_3, \ldots, z_n)$ is a basis: applying φ to a zero combination forces the coefficient of w to vanish; the remaining coefficients vanish by independence in the kernel. Now $Au = Az_j = 0$ and $Aw = u$, giving the matrix E_{12}. Two rank-one matrices with the same size and trace are therefore similar to the same displayed matrix, hence to one another.

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